KF Parameter Prediction

from Frequency-Stability Curves

Instant initial guesses for Kalman-filter tuning on atomic clock data.

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The problem

Kalman filters need good q-values. Getting them is slow.

Use case

"I have clock phase measurements.
I don't know the q values.
I want to optimize the Kalman filter
— give me a quick initial guess."

Current options:

- NLL optimization — multi-minute, highly sensitive to warm start
- ALS — well-conditioned but iterative, needs seed
- Manual MHDEV fit — requires expert τ-region selection

Research question

Can supervised learning on frequency-stability features predict KF q-parameters accurately enough to seed or replace iterative optimizers?

Sub-question

Which stability statistic is most informative — ADEV, MDEV, HDEV, or MHDEV?

Targets: $\log_{10}(q0)$, $\log_{10}(q1)$, $\log_{10}(q2)$

Why it matters

Clocks are everywhere — but their noise models aren't automatic.

Atomic clocks drift

GPS, telecom, defense timing, scientific instrumentation. Operators get phase data; they want stability predictions.

KF tuning is slow

NLL optimization: 10–60 s per sample, sensitive to warm start. Lab workflow bottleneck.

ML can do it in ms

Trained regressor: sub-millisecond inference on raw features. Quantified uncertainty included.

Dataset pipeline

10,000 synthetic samples anchored to real GMR6000 Rb data.

Real GMR6000 Rb phase
35 days, 1 PPS

→

h-range sampling
Kasdin 1992

→

Composite noise gen.
N = 524,288 (~6 days)

→

196-feature extraction
ADEV, MDEV, HDEV, MHDEV

→

ALS label optimizer
noise-free vs truth

Key design decisions

Physically grounded

h_α ranges anchored to measured GMR6000 Rb (not arbitrary noise blends).

4+1 noise types

WPM + WFM + FFM + RWFM always; FPM in ~30% of samples.

Targets are $\log_{10}(q)$

Log-space equalizes multi-decade dynamic range; KF-native.

ALS-optimized labels

Probed on 10 draws: ALS matches truth to $\leq 10^{-3}$ dec; NLL drifted ±8 dec.

Feature engineering

196 features per sample from 20 log-spaced τ values.

Feature decomposition

Group	Count	Description
Raw σ values	80	4 stats × 20 τ: $\log_{10}$ of ADEV, MDEV, HDEV, MHDEV
Log-log slopes	76	4 stats × 19 adjacent-τ slopes (noise-type signature)
Variance ratios	40	MVAR/AVAR (20) + MHVAR/HVAR (20) — bias-type detectors
Total	**196**	**Per-sample feature vector**

Why these four statistics?

ADEV — standard reference; baseline comparison

MDEV — distinguishes WPM vs FPM (slope −3 vs −2)

HDEV — Hadamard; rejects frequency drift

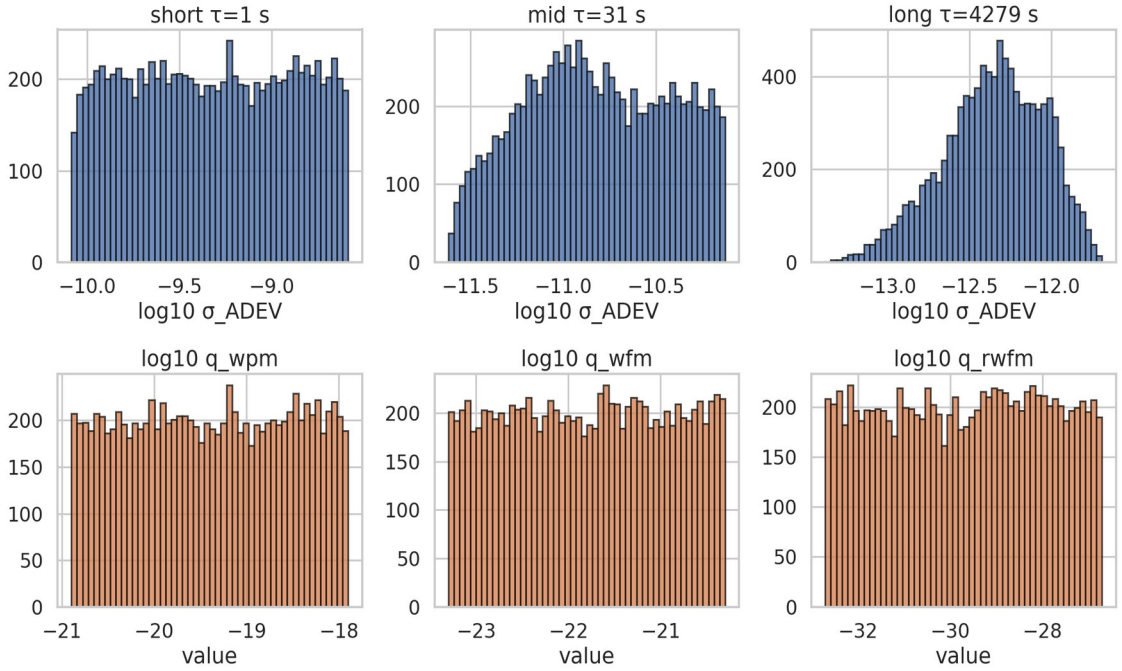
MHDEV — modified Hadamard; best for long τ on drifting clocks

Secondary research hypothesis

Feature-importance analysis will reveal which statistic carries the most information about KF noise parameters.

Data exploration

Target distributions, example σ(τ) curves, feature–target correlations.



What the data shows

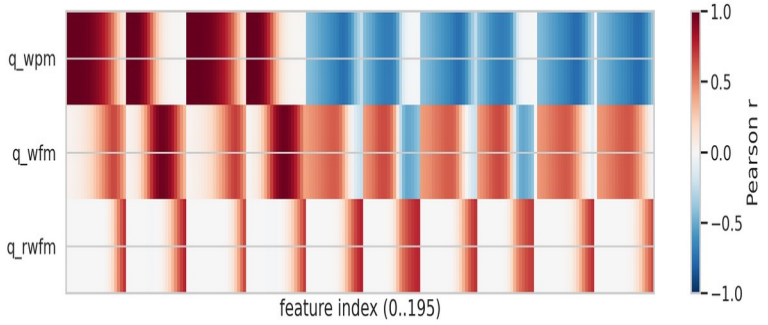
- Targets span ~6 log decades (q2 widest; q0 tightest)
- Short-τ features (top row) correlate strongest with q0
- Long-τ features correlate with q2 ($\log_{10} q \sim -30$)
- FPM-present samples (30%) have steeper short-τ slope
- NaN rate 0%; no outliers at $|z|>5$ — clean dataset

Statistical baselines

Setting the bar: what does 'good' look like?

Baseline predictors

Method	RMSE ($\log_{10}$ q)	Notes
Naive (train mean)	0.87 / 0.87 / 1.73	Ignores features ($R^2 \approx 0$)
MHDEV 3-region fit	see slide 13	Real-data only, requires expert τ-regions
RF (ours)	**0.05 / 0.08 / 0.79**	**~18×/11×/2× vs naive**



Correlation pattern above: short-τ features drive q0; long-τ features drive q2. *Relationships are monotonic but clearly nonlinear — motivates tree ensembles.*

Methodology

Random Forest + XGBoost, multi-output regression.

Random Forest

- Bagged ensemble of decision trees
- Built-in UQ via tree-variance (forestci)
- Invariant to monotone feature transforms
- Robust baseline for tabular data

XGBoost (Gradient Boosted)

- Sequential boosting — corrects errors
- Typically ~1–3× better RMSE than RF on tabular
- UQ via quantile regression (α=0.05, 0.95)
- Best-in-class for multi-output regression

Why both?

Head-to-head comparison on identical features. Different bias/variance tradeoffs → diagnostic of whether signal is smooth (boosting wins) or rough (RF wins). UQ mechanisms differ — cross-check coverage.

Hyperparameter tuning

Two-pass 5-fold GridSearchCV: expand, then re-center around boundary winners.

Random Forest grid

Parameter	Grid	Best
n_estimators	{1000, 1500, 2000}	1000
max_depth	{20}	20
min_samples_leaf	{1, 2, 3}	1
max_features	{0.5}	0.5

XGBoost grid

Parameter	Grid	Best
n_estimators	{500, 750, 1000}	750
learning_rate	{0.003, 0.01}	0.01
max_depth	{6}	6
subsample	{0.7, 0.8}	0.7

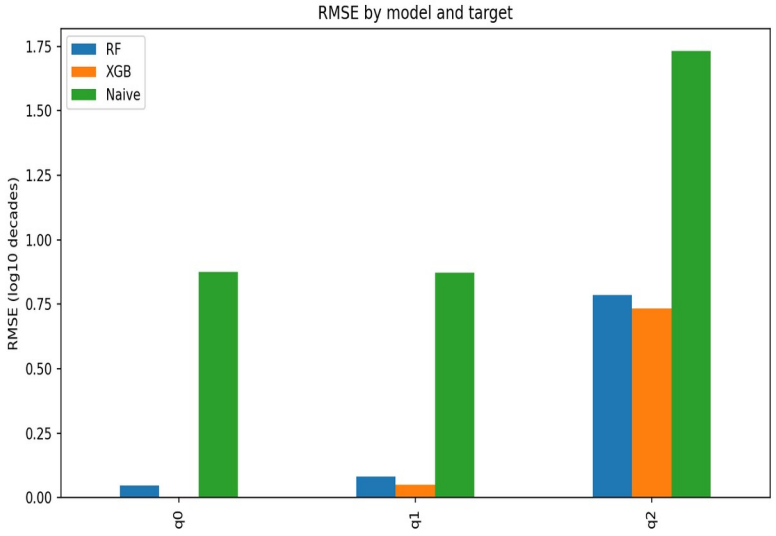
CV protocol

5-fold CV • Pass 1 landed on boundaries (n_est top, lr bottom, subsample bottom) • **Pass 2 probe** = 45 RF + 60 XGB fits centered on pass-1 winners, extending past boundaries → **new winners are interior**; scoring = neg-MSE in $\log_{10}$-q; seed = 42.

Results

Per-target metrics + model comparison.

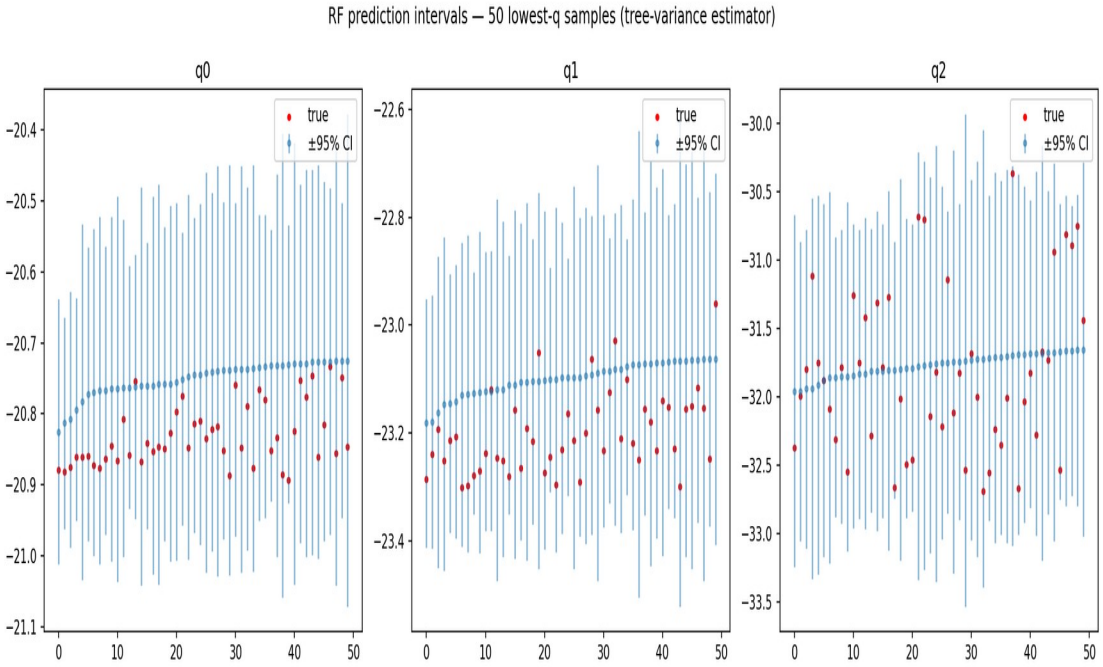
Target	Model	RMSE	R^2	MAE
q0	Naive	0.874	0.000	0.759
q0	RF	0.047	0.997	0.034
q0	**XGB**	**0.004**	**0.99998**	**0.003**
q1	Naive	0.873	0.000	0.758
q1	RF	0.080	0.992	0.055
q1	**XGB**	**0.048**	**0.997**	**0.027**
q2	Naive	1.732	0.000	1.494
q2	RF	0.786	0.794	0.552



Headline: XGB → <0.05-dec RMSE on q0 & q1 ($R^2 \geq 0.997$); **q2 R^2 = 0.82** (vs 0.65 before switching labels from NLL → ALS).

Uncertainty quantification

Grad-student requirement — the model knows when it doesn't know.



Two complementary approaches

Random Forest — forestci
Infinitesimal jackknife on tree predictions
→ *analytic variance*

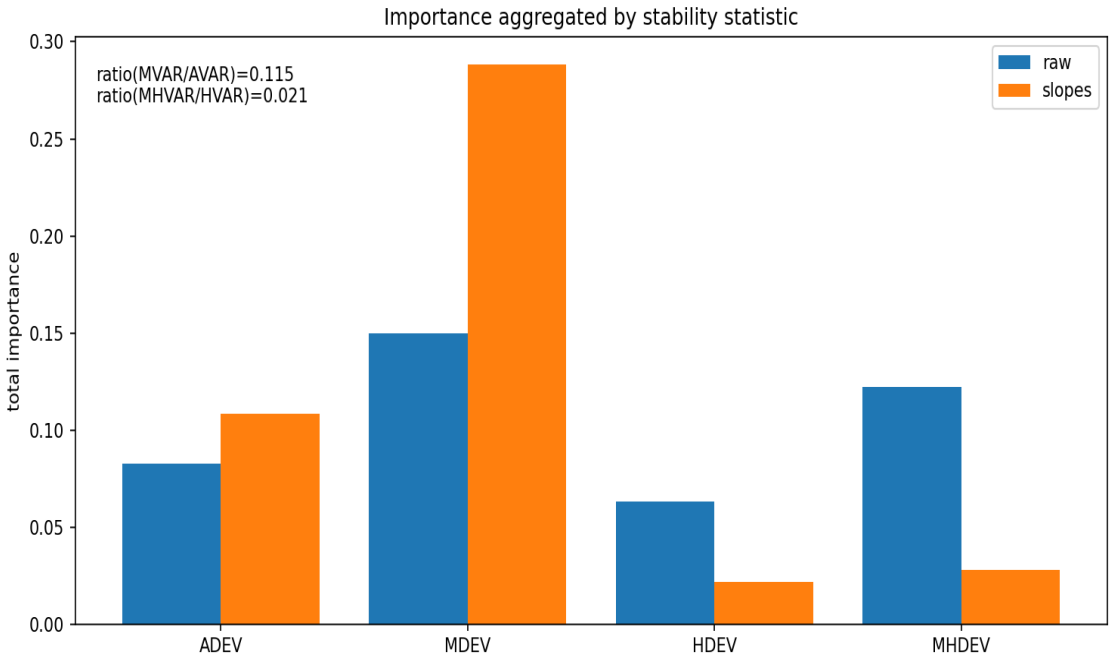
XGBoost — quantile regression
Two extra models at α=0.05, α=0.95
→ *non-parametric prediction intervals*

Empirical 90% PI coverage (XGB):

q0 84% q1 81% q2 81%

Which statistic carries the signal?

Headline: feature-importance answer to the sub-question.



What the data says

MDEV dominates (Σ importance ≈ 0.59)

MHDEV 5× less important (Σ ≈ 0.12)

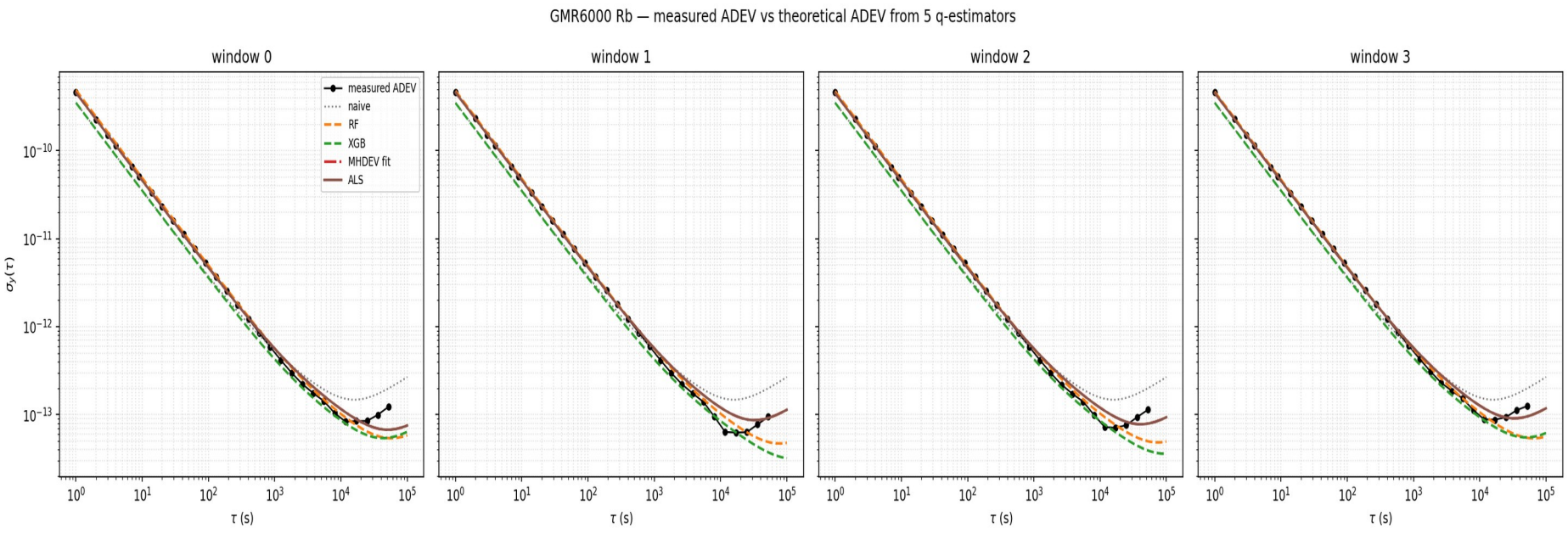
ADEV / HDEV contribute the remaining ~0.29

Slopes carry more signal than raw σ values — *consistent with noise-type diagnostics driving q.*

Contribution: *MDEV is the most informative stability statistic for KF tuning.*

Real-data validation

GMR6000 Rb phase → ML prediction → ADEV overlay on 4 windows.



Headline: RF & XGB σ(τ) predictions track measured ADEV across all 4 windows — **model generalizes from synthetic to real hardware.**

Takeaways & limitations

What worked

- ALS labels → noise-free targets (key finding; lifted q2 R^2 from 0.65 → 0.82)
- XGB → 0.004-dec RMSE on q0 (R^2 = 0.99998); <0.05-dec on q1
- forestci + quantile UQ → 80–84% empirical coverage of nominal 90% PIs
- Feature importance → MDEV quantitatively most informative (5× MHDEV)

Limitations

- Trained on Rb GMR6000-class h-ranges only
- 3-state KF model lacks flicker-FM — limits long-horizon holdover
- Single real-data record (6k27feb) for cross-check
- XGB 90% PI under-covers by 6–10 pp (conservative regularization)

Future work

- Extend to OCXO, HSO, cesium training ranges
- 4-state model with explicit FFM for sub-ns long-term holdover
- End-to-end NN on raw phase (current: hand-crafted features)
- Active learning on where the model is least confident

Thank you

Questions?

Code
github.com/ianlap/SigmaTau

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